

Model Report

1. Approach & Architecture

Our system packages the pytorch model exported via ONNX Runtime.

- **Preprocessing Pipeline:** Input images are resized to 32×32 pixels, converted to RGB, and normalized to the $[0, 1]$ range.
- **Data Format:** The system adheres to the standard NCHW (Batch, Channel, Height, Width) channel-first convention, ensuring direct compatibility with our PyTorch-trained model tensors.

2. Validation Rigor

To ensure reliability and reproducibility, we implemented the following validation measures:

- **Preprocessing Verification:** Automated unit tests confirm that all incoming tensors strictly adhere to the (1, 3, 32, 32) shape and standard floating-point normalization.
- **Contract Integrity:** We enforce strict schema validation via Pydantic. Any output falling outside the probability range of $[0.0, 1.0]$ is caught by a handler and clamped into the correct range.
- **Security & Safety Protocols:**
 - **SSRF Protection:** We resolve hostnames to IP addresses and block private/loopback ranges to prevent the API from probing internal network infrastructure.
 - **Resource Throttling:** Strict file size and type enforcement (max 5MB) reduces risk from oversized uploads and even prevents Denial of Service (DoS) attacks.

3. Feature Choices and Trade-offs

Feature Choice	Justification	Trade-off / Impact
Probability Thresholds (0.33/0.66)	Categorizes predictions into 'Fake', 'Unsure', and 'Real' buckets to manage model uncertainty.	Positive: Provides transparency by flagging ambiguous images as uncertain rather than forcing a potentially incorrect binary label.
CPU Execution	Ensures total portability across judge environments.	Negative: Inference latency is higher compared to GPU-optimized providers.
Minimal Augmentation	Ensures consistent, reproducible results.	Negative: Reduces model robustness against extreme image orientation or noise variations.

4. Limitations

The model is constrained by the training dataset’s limited diversity. Because the model was trained on fixed-resolution 32x32 images, it lacks the capacity to analyze high-frequency textures found in modern generative media, which may result in lower accuracy on real-world, high-resolution web imagery.